

Decision Trees

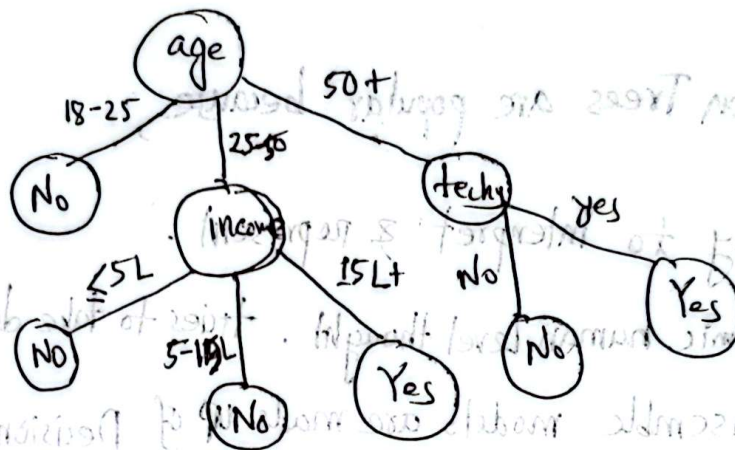
1. Decision Trees are supervised models.
2. Can be used for both classification/regression task.
3. They are simple tree like structure (hierarchical in nature)
4. Decision trees can be thought as nested if else conditions.
5. Highly interpretable models, easy to explain the working.

Decision Trees are popular because

- i. Easy to interpret & represent.
- ii. Mimic human level thought. tries to take decisions like a human does
- iii. Ensemble models are made up of Decision trees that performs even better than individual Decision trees.
- iv. When feature are categorical, Decision Trees are preferred over other models.

Ex: Solve a business problem that wants to predict whether a user clicks on an advertisement of car or not?

Sex	Income	alreadyCar	Techy	Age
M	$\leq 5L$	Yes	Yes	18-25
F	5L-15L	No	No	25-50
	15L+			50+



if age betⁿ 18-25:
return No

if age betⁿ 25-50:
if income $\leq 15L$:
return No
else return Yes

if age 50+:
if techy:
return Yes
else return No

Ex: $X_{Test} = \langle M, 18L, No, Yes, 35 \rangle$

Ans: Yes

But Question is that,
 ↪ Which feature comes to root and which comes to leaf?
 ↪ Who decide?
 ↪ Which feature need to eliminate?

Information Gain

Entropy

method

Entropy: Entropy is measure of randomness of a system.

$$H(x) = - \sum_{i=1}^n P_i \log_2 P_i$$

n = different unique value

Ex

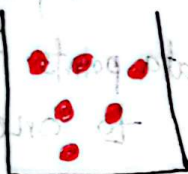
Case-I



$$H(x) = - \sum_{i=1}^n P_i \log_2 P_i$$

$$= - \left[\underbrace{\frac{3}{6} \log_2 \frac{3}{6}}_{\text{Red}} + \underbrace{\frac{3}{6} \log_2 \frac{3}{6}}_{\text{Green}} \right]$$

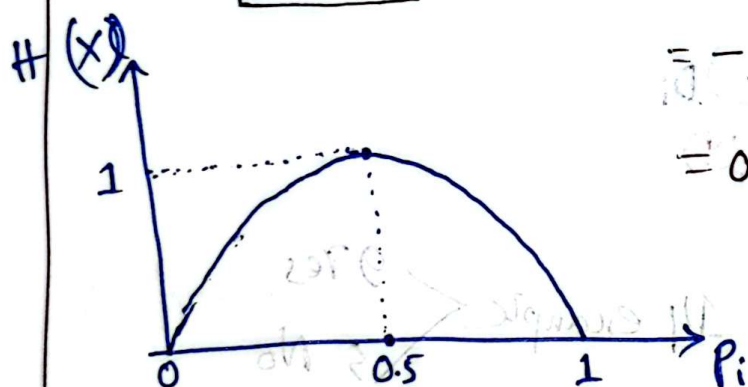
Case-II



$$H(x) = - \sum_{i=1}^n P_i \log_2 P_i$$

$$= - [1 \log_2 \frac{1}{2}]$$

$$= 0.918$$



When probability is equal for all classes, then Entropy is maximum.

In this example, class $\rightarrow 2$

$$\text{So, } \frac{1}{2} = 0.5$$

At point 0.5 the Entropy is maximum

Information Gain: Information gain quantifies how much a feature helps in reducing uncertainty (entropy) when splitting the data.

$$\text{Gain}(Y, A) = H_0(Y) - \sum_{i=1}^K \left(\frac{|D_i|}{|D|} \right) H_{D_i}(Y)$$

Information Gain of system Y

When split by attribute A

Entropy of original system (Parent)

Weight factor

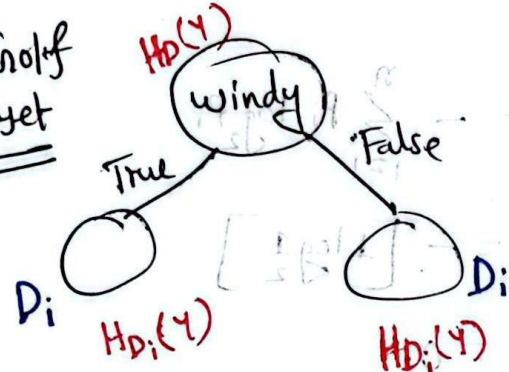
Entropy of smaller Node

K = How many splits are doing

D = Total Datapoint in the parent

D_i = Data points that belongs to one Node.

For Golf Dataset



For Golf Dataset (GitHub)

14 example $\begin{cases} 9 \text{ Yes} \\ 5 \text{ No} \end{cases}$

$$\text{Entropy}, H(S) = - \left[\frac{9}{14} \log_2 \frac{9}{14} + \frac{5}{14} \log_2 \frac{5}{14} \right]$$

$$= 0.94$$

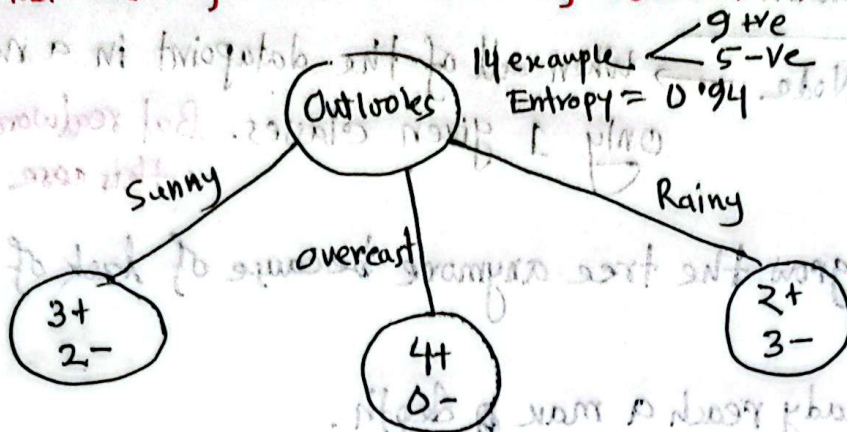
Play Golf Dataset

	OUTLOOK	TEMPERATURE	HUMIDITY	WINDY	PLAY GOLF
0	Rainy	Hot	High	False	No
1	Rainy	Hot	High	True	No
2	Overcast	Hot	High	False	Yes
3	Sunny	Mild	High	False	Yes
4	Sunny	Cool	Normal	False	Yes
5	Sunny	Cool	Normal	True	No
6	Overcast	Cool	Normal	True	Yes
7	Rainy	Mild	High	False	No
8	Rainy	Cool	Normal	False	Yes
9	Sunny	Mild	Normal	False	Yes
10	Rainy	Mild	Normal	True	Yes
11	Overcast	Mild	High	True	Yes
12	Overcast	Hot	Normal	False	Yes
13	Sunny	Mild	High	True	No



Golf

Q Find the Information Gain of Dataset if split by Outlook.



$$I.G(D, outlook) = \underbrace{H(Y)}_{\text{outlook system Entropy} = 0.94} - \sum_{i=1}^K \frac{|D_i|}{|D|} H_{D_i}(Y)$$

$$= 0.94 - \left[\underbrace{\left(-\left(\frac{3}{5} \log_2 \frac{3}{5} + \frac{2}{5} \log_2 \frac{2}{5} \right) \right) \frac{5}{14}}_{\text{Sunny}} - \underbrace{\left(\frac{4}{4} \log_2 \frac{4}{4} + \frac{0}{4} \log_2 \frac{0}{4} \right) \frac{4}{14}}_{\text{Overcast}} - \underbrace{\left(\frac{2}{5} \log_2 \frac{2}{5} + \frac{3}{5} \log_2 \frac{3}{5} \right) \frac{5}{14}}_{\text{Rainy}} \right]$$

$$= 0.27$$

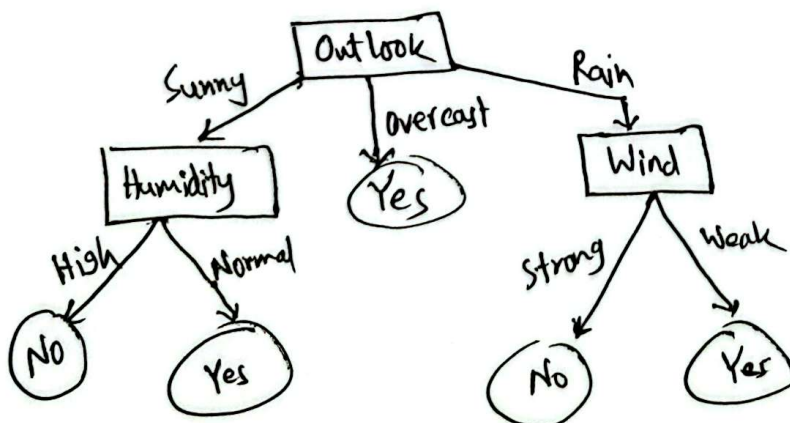
Similarly by coding we get,

$$I.G(D, Humidity) = 0.15$$

$$I.G(D, Windy) = 0.048$$

$$I.G(D, Temperature) = 0.0292$$

Constructing a Decision Tree



Recursively call

But Question is,
When we stop?

Next page details

Stopping Condition

- case-I ① Pure Node → When all of the datapoint in a node has only 1 given classes. *But realworld dataset this case is rare.*
- case-II ② Can't grow the tree anymore because of lack of points.
- ③ If already reach a max depth.

$$H(D) = - \sum_{i=1}^n p_i \log_2 p_i$$

$$H(D) = - \left(\frac{1}{2} \log_2 \frac{1}{2} + \frac{1}{2} \log_2 \frac{1}{2} \right) = - \left(\frac{1}{2} (-1) + \frac{1}{2} (-1) \right) = 1$$

$$H(D) = 1$$

$$H(D) = 1$$

$$H(D) = 1$$

